

60. Two points P and Q lie on line $y = 2x + 3$. These two points P and Q are at a distance 2 units from another point $R(1, 5)$. What are the coordinates of the points P and Q ?
- (a) $\left(1 + \frac{2}{\sqrt{5}}, 5 + \frac{4}{\sqrt{5}}\right), \left(1 - \frac{2}{\sqrt{5}}, 5 - \frac{4}{\sqrt{5}}\right)$
 (b) $\left(3 + \frac{2}{\sqrt{5}}, 5 + \frac{4}{\sqrt{5}}\right), \left(-1 - \frac{2}{\sqrt{5}}, 5 - \frac{4}{\sqrt{5}}\right)$
 (c) $\left(1 - \frac{2}{\sqrt{5}}, 5 + \frac{4}{\sqrt{5}}\right), \left(1 + \frac{2}{\sqrt{5}}, 5 - \frac{4}{\sqrt{5}}\right)$
 (d) $\left(3 - \frac{2}{\sqrt{5}}, 5 + \frac{4}{\sqrt{5}}\right), \left(-1 + \frac{2}{\sqrt{5}}, 5 - \frac{4}{\sqrt{5}}\right)$
61. If two sides of a square lie on the lines $2x + y - 3 = 0$ and $4x + 2y + 5 = 0$, then what is the area of the square in square units?
 (a) 6.05 (b) 6.15 (c) 6.25 (d) 6.35
62. ABC is a triangle with $A(3, 5)$. The mid-points of sides AB, AC are at $(-1, 2), (6, 4)$ respectively. What are the coordinates of centroid of the triangle ABC ?
- (a) $\left(\frac{8}{3}, \frac{11}{3}\right)$ (b) $\left(\frac{7}{3}, \frac{7}{3}\right)$
 (c) $\left(2, \frac{8}{3}\right)$ (d) $\left(\frac{8}{3}, 2\right)$
63. ABC is an acute angled isosceles triangle. Two equal sides AB and AC lie on the lines $7x - y - 3 = 0$ and $x + y - 5 = 0$. If θ is one of the equal angles, then what is $\cot \theta$ equal to?
 (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) 2
64. In the parabola $y^2 = 8x$, the focal distance of a point P lying on it is 8 units. Which of the following statements is/are correct?
- The coordinates of P can be $(6, 4\sqrt{3})$.
 - The perpendicular distance of P from the directrix of parabola is 8 units.
- Select the correct answer using the code given below:
 (a) 1 only (b) 2 only
 (c) Both 1 and 2 (d) Neither 1 nor 2
65. What is the eccentricity of the ellipse if the angle between the straight lines joining the foci to an extremity of the minor axis is 90° ?
- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{1}{\sqrt{3}}$ (d) $\frac{1}{\sqrt{2}}$
66. Let $\vec{a} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$. If $\vec{a} \times (\vec{b} \times \vec{a}) = \alpha \hat{i} - \beta \hat{j} + \gamma \hat{k}$, then what is the value of $\alpha + \beta + \gamma$?
 (a) 8 (b) 7 (c) 6 (d) 0
67. If a vector of magnitude 2 units makes an angle $\frac{\pi}{3}$ with $2\hat{i}$, $\frac{\pi}{4}$ with $3\hat{j}$ and an acute angle θ with $4\hat{k}$, then what are the components of the vector?
 (a) $(1, \sqrt{2}, 1)$ (b) $(1, -\sqrt{2}, 1)$
 (c) $(1, -\sqrt{2}, -1)$ (d) $(1, \sqrt{2}, -1)$
68. Consider the following in respect of moment of a force:
- The moment of force about a point is independent of point of application of force.
 - The moment of a force about a line is a vector quantity.
- Which of the statements given above is/are correct?
 (a) 1 only (b) 2 only
 (c) Both 1 and 2 (d) Neither 1 nor 2
69. For any vector $\vec{r}$, what is $(\vec{r} \cdot \hat{i})(\vec{r} \times \hat{i}) + (\vec{r} \cdot \hat{j})(\vec{r} \times \hat{j}) + (\vec{r} \cdot \hat{k})(\vec{r} \times \hat{k})$ equal to?
 (a) $\vec{0}$ (b) $\vec{r}$ (c) $2\vec{r}$ (d) $3\vec{r}$
70. Let $\vec{a}$ and $\vec{b}$ are two vectors of magnitude 4 inclined at an angle $\frac{\pi}{3}$, then what is the angle between $\vec{a}$ and $\vec{a} - \vec{b}$?
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$
71. Let $y_1(x)$ and $y_2(x)$ be two solutions of the differential equation $\frac{dy}{dx} = x$. If $y_1(0) = 0$ and $y_2(0) = 4$, then what is the number of points of intersection of the curves $y_1(x)$ and $y_2(x)$?
 (a) No point
 (b) One point
 (c) Two points
 (d) More than two points
72. The differential equation, representing the curve $y = e^x(a \cos x + b \sin x)$ where a and b are arbitrary constants, is